Notes

Introduction

as the objects in our everyday lives do: Werner Heisenberg 1958, *Physics and Philosophy*, Harper Torchbooks ed. (Harper and Row), p. 129.

"shut up and calculate": N. David Mermin 1990, *Boojums All the Way Through: Communicating Science in a Prosaic Age* (Cambridge), p. 199.

"The theory decides what we can observe": Werner Heisenberg 1971, *Physics and Beyond* (HarperCollins), p. 63.

three centuries after Galileo's discovery: See Stanley L. Jaki 1978, "Johann Georg von Soldner and the Gravitational Bending of Light, with an English Translation of His Essay on It Published in 1801," *Foundations of Physics* 8 (11/12): 927–950. The experiment would have been possible decades before Einstein—and was in fact proposed a century before Einstein, by Johann Soldner, as a test of Newtonian physics. But nobody cared until Einstein proposed a rival theory to Newtonian gravity that could be tested in this manner.

Prologue

"but I *knew* it was *rotten*": Jeremy Bernstein 1991, *Quantum Profiles* (Princeton University Press), p. 20. Emphasis on "knew" is Bell's, according to Bernstein; emphasis on "rotten" is inferred from the context in Bernstein: "Bell pronounced the word 'rotten' with a good deal of relish."

"got on with more practical things": John S. Bell 2004, *Speakable and Unspeakable in Quantum Mechanics*, 2nd ed. (Cambridge University Press), p. 160.

"the impossible done": Bell 2004, p. 160.

"waiting for me": Charles Mann and Robert Crease 1988, "Interview: John